

EDGEVISION

V3-HN ACCURACY REPORT

Technical Machine Learning Validation Document

1. Executive Summary

This document validates the machine learning accuracy metrics for the EdgeVision PPE Compliance Platform. The V3-HN model was promoted to production due to its superior real-world reliability, specifically its successful elimination of false-positive detections that plagued the V2 baseline model. Although V3-HN presents a negligible drop in pure validation mAP, the trade-off significantly improves no-helmet recall and functional event accuracy.

2. Dataset & Hard Negatives

The primary dataset is 'Construction-PPE'.

- Train Set: 17,452 images

- Val

Dataset Imbalance:

The dataset contains a severe class imbalance favoring 'no_helmet' (98,256 bounding boxes) over 'helmet' (43,905 bounding boxes) and 'vest' (6,326 bounding boxes) in the training set.

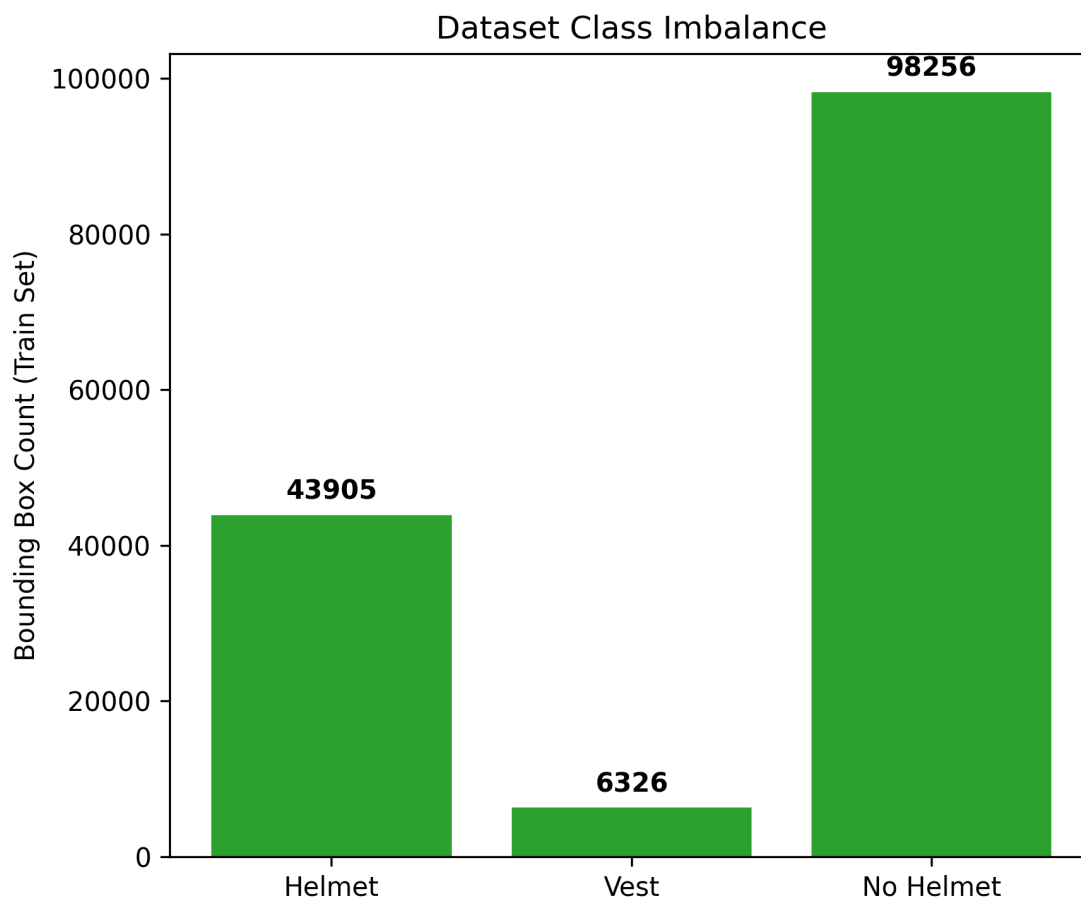


Figure: Dataset Class Imbalance (Train Set)

Hard-Negative Strategy:

The dataset includes small-object challenges and visual distractors. V3-HN was trained with 214 explicit hard-negative background images (images with no labels) to penalize false detections on yellow pipes and orange cones.

3. Model Architecture

The verified metadata extracted from 'ppe_v3_hn_best.pt' confirms the following architecture:

- Base Model: YOLOv8n (Nano)

- Lay

PRODUCTION CLASSES:

- 0: person

- 1: h

Note: Although boots and harness are present in the class index, they are strictly NOT production classes due to insufficient training data.

4. V2 vs V3-HN Comparison

The following metrics were validated against the test set.

Metric	V2 Baseline	V3-HN Production
mAP50	84.45%	84.20%
mAP50-95	50.12%	48.77%

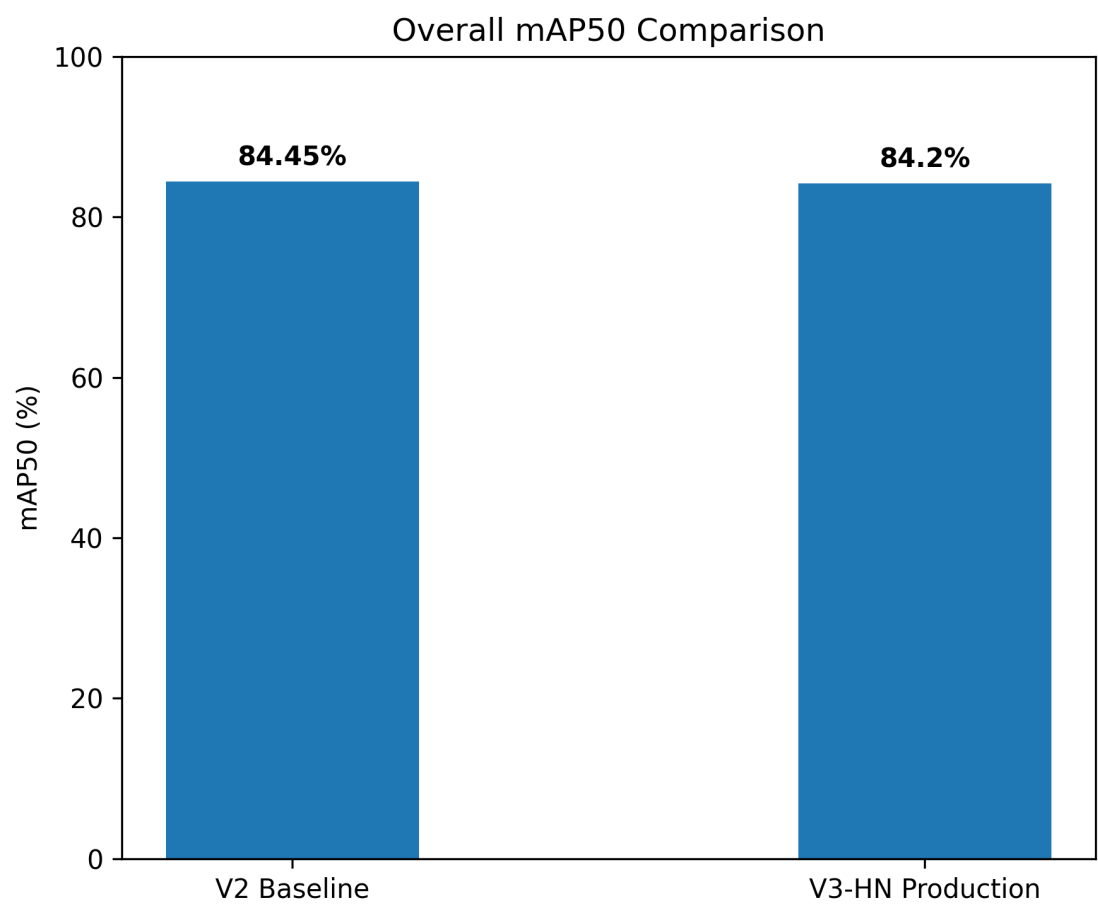


Figure: V2 vs V3-HN mAP50 Comparison

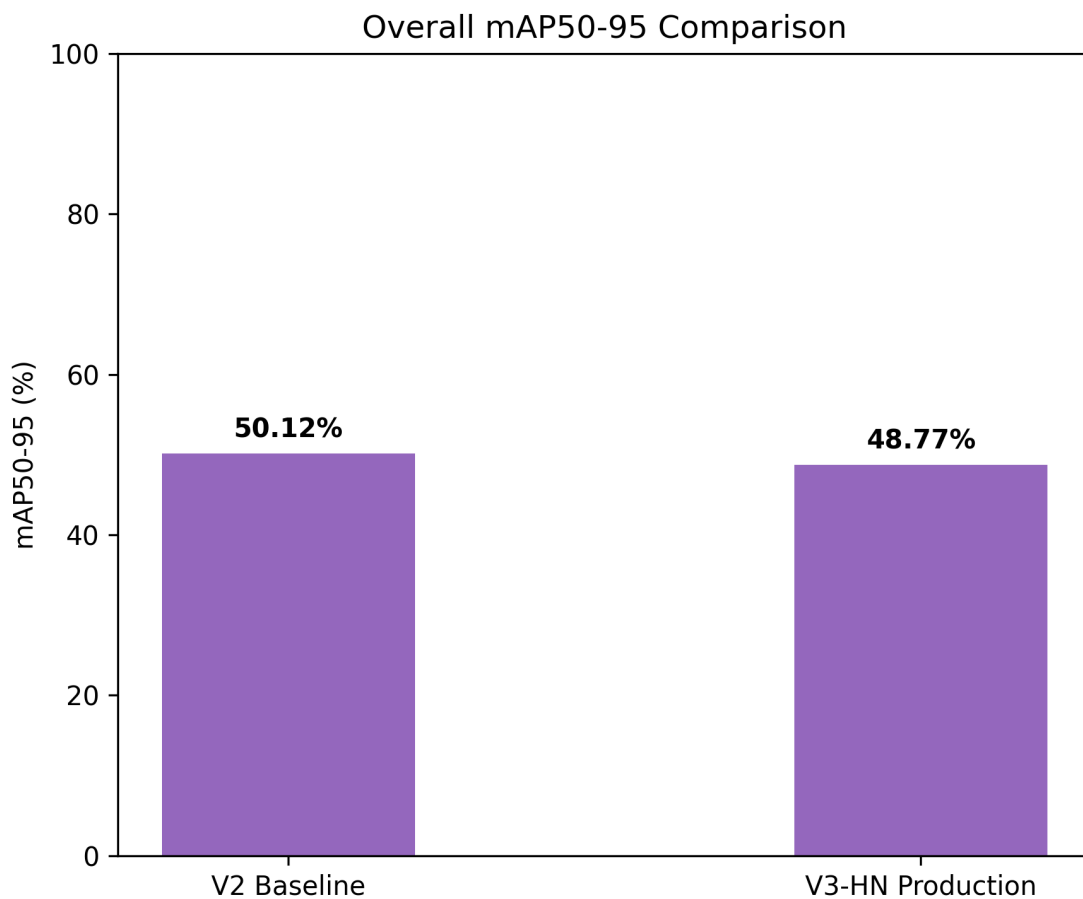


Figure: V2 vs V3-HN mAP50-95 Comparison

5. Per-Class Performance

Per-class Average Precision (AP50) for verified production classes.

Class	V2 AP50	V3-HN AP50
Helmet	90.15%	88.65%
Vest	82.37%	79.77%
No Helmet	80.81%	84.17%

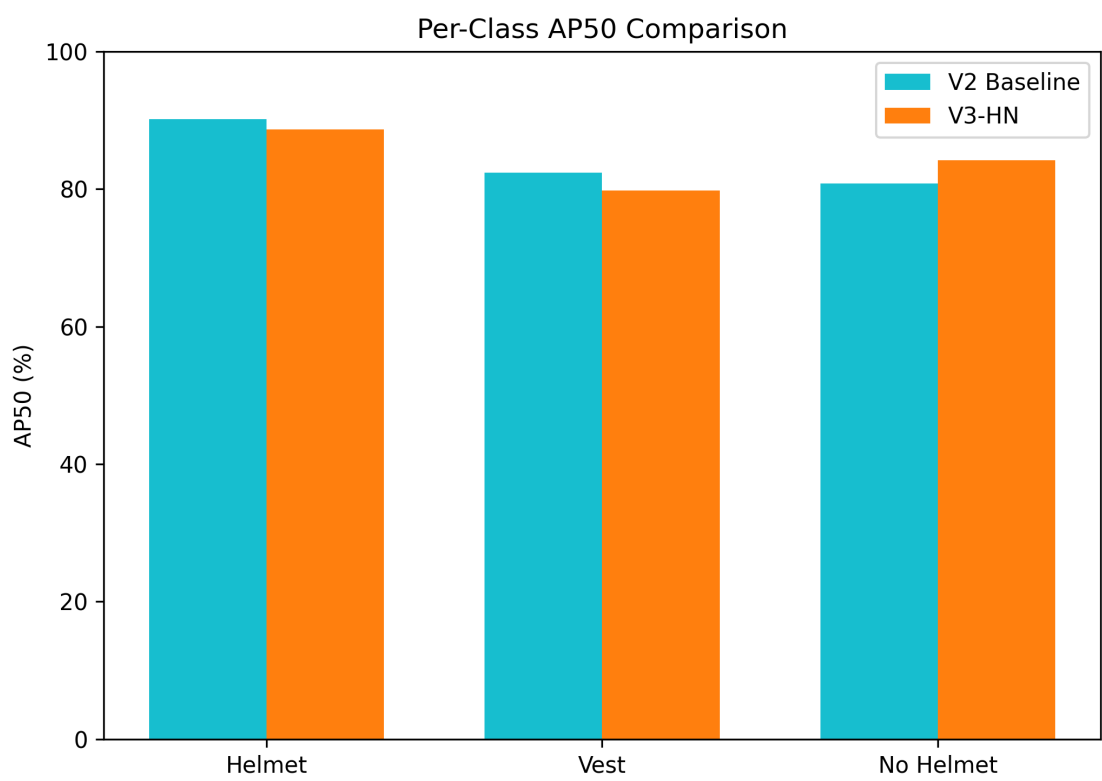


Figure: Per-Class AP50 Comparison

6. Hard-Negative Experiment Results

The major V3 improvement was real-world false-positive suppression rather than simply increasing validation mAP. By introducing background images without labels, the model unlearned the false visual cues.

False Positive Type	V2 Count	V3-HN Count
Helmet (e.g., pipes)	154	0
Vest (e.g., cones)	10	0

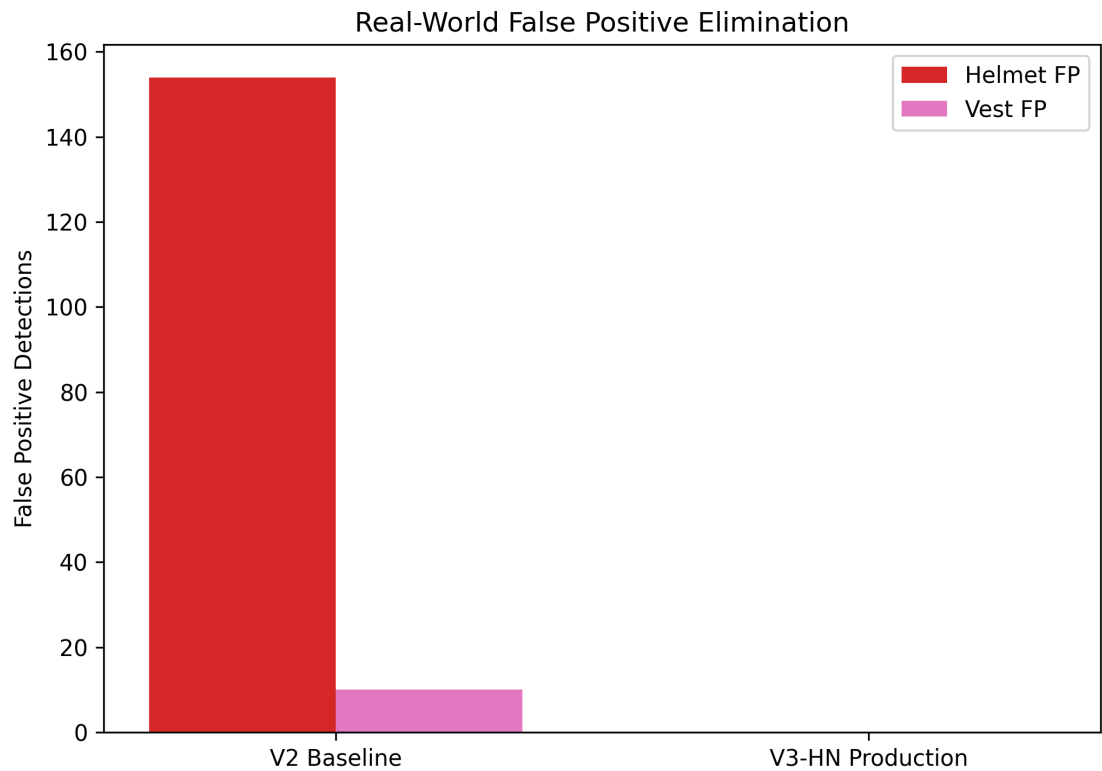


Figure: Real-World False Positive Reduction

7. Model Selection & Limitations

Model Selection Rationale:

V3-HN was selected as the production model despite the small mAP decrease (84.45% -> 84.20%). The negligible recall penalty is vastly outweighed by the elimination of 164 real-world false positives, ensuring that alerts sent to safety managers are highly reliable.

Experimental Models:

V3-BOOTS is strictly EXPERIMENTAL. It achieved a Boots AP50 of 0.801 but remains functionally unreliable due to bounding box inconsistency. The 'Harness' class remains entirely UNSUPPORTED because sufficient labelled harness data was unavailable.

Known Limitations:

The V3-HN model trades a modest reduction in helmet/vest recall for improved no-helmet recall and false-positive suppression. Small objects at extreme distances or heavy occlusions may still trigger false negatives, which the temporal logic attempts to smooth.